IK 502 - Introduction to Biosignals

Assignment: PPG-Derived Heart Rate Variability (HRV) Data Processing and Analysis

Signal processing using a reproducible script

1. Objective

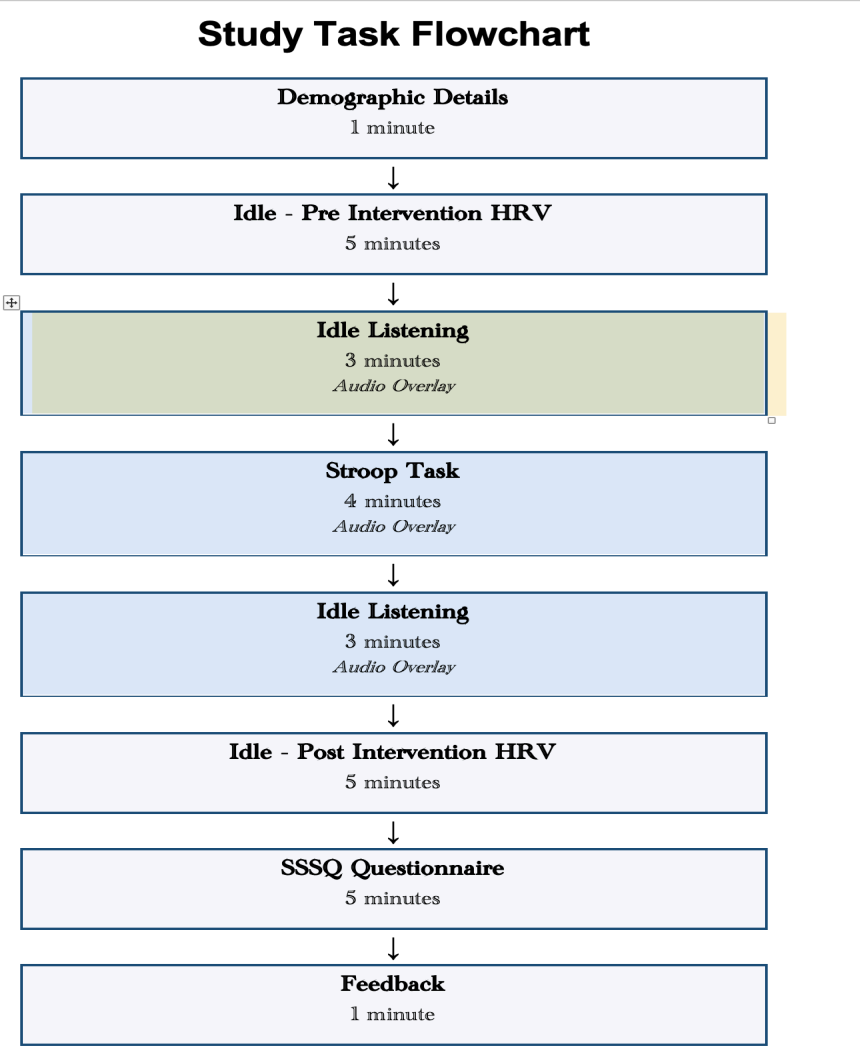
Process the provided PG sensor data using a script and perform heart rate variability (HRV) analysis across the three study phases: Pre-Intervention, During Intervention, and Post-Intervention.

2. Study Protocol

Total study duration for the HRV analysis: 20 minutes.

Study Phase	Activity	Duration
Pre-Intervention	Idle / resting recording	5 minutes
During Intervention	Listening	3 minutes
During Intervention	Task while listening	4 minutes
During Intervention	Listening	3 minutes
Post-Intervention	Idle / resting recording	5 minutes

Intervention duration: 10 minutes (3-minute listening + 4-minute task while listening + 3-minute listening).



Study task flowchart

3. Data Analysis Assignment

Students are required to write and use a script for processing the provided PG sensor recording. The script should separate the recording according to the study protocol and calculate the specified HRV metrics for comparison across the Pre-Intervention, During-Intervention, and Post-Intervention sessions.

4. Required HRV Metrics

Calculate the following metrics:

- Time-domain: SDNN
- Time-domain: RMSSD
- Frequency-domain: LF power (0.04–0.15 Hz)
- Frequency-domain: HF power (0.15–0.40 Hz)

5. Analysis Conditions

The four HRV metrics must be obtained for the following 3 conditions:

1. Pre-Intervention session — **5 minutes**

2. During Intervention — **10 minutes** in total

   • Listening — first 3 minutes

   • Task while listening — 4 minutes

   • Listening — final 3 minutes

3. Post-Intervention session — **5 minutes**

6. Required Comparison

Compare the calculated HRV metrics across Pre-Intervention, During-Intervention, and Post-Intervention sessions. The three components of the intervention period should first be identified separately in the script so that the intervention data are correctly segmented according to the study protocol.

7. Script Requirement

The complete data processing and HRV analysis must be performed using a reproducible script. The script should include data import, segmentation according to the study protocol, signal processing, pulse/peak detection, interval extraction, HRV metric calculation, and comparison across the required study phases.

8. Submission

1. Analysis script used for processing the provided data.

2. Calculate the four HRV metrics

HRV Metric	Pre	During	Post
SDNN (ms)			
RMSSD (ms)			
LF Power (ms^2)			

HRV Metric	Pre	During	Post
HF Power (ms^2)			

3. Calculate percentage change

- Pre → During % change
- During → Post % change

for **SDNN, RMSSD, LF power, and HF power**.

4. **Plots:**

Generate appropriate plots for the Pre-Intervention, During-Intervention, and Post-Intervention phases to show:

1. Heart rate or beat-to-beat interval variation over time.
2. Comparison of SDNN and RMSSD.
3. Power spectral density (PSD), indicating the LF and HF frequency bands.
4. Comparison of LF and HF power.

Instructions for Software/Tools

Students may use MATLAB, Python, or another suitable scientific signal-processing environment to complete the assignment. The analysis should be performed using a script/code-based workflow so that all processing steps are reproducible.

Platform	Suggested tools/libraries
MATLAB	Signal Processing Toolbox
Python	NumPy, Pandas, SciPy, Matplotlib
Python	NeuroKit2, MNE (optional)

- ***Students are not restricted to a particular programming language or package. However, the submitted work must demonstrate the processing of the provided PG signal and calculation of the required HRV parameters.***

  Drive link for the files

- ***https://drive.google.com/drive/folders/1mwrnuwVBAfaCmRNB4fFTNAkrY4h5TdM3?usp=drive_link***